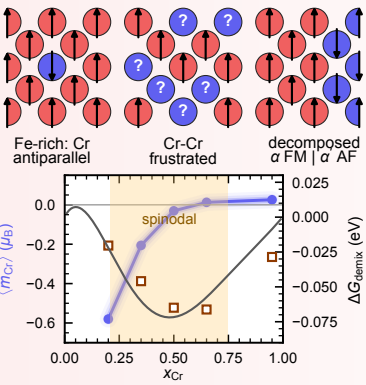
FRUSTRATE

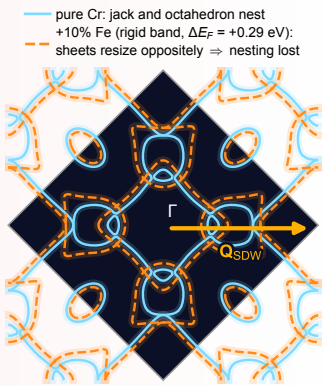
(a) atoms & moments

schematic



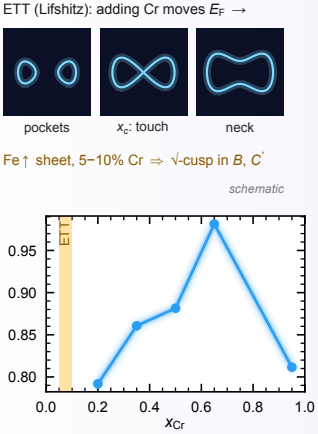
NEST

(b) Cr Fermi surface vs Fe content



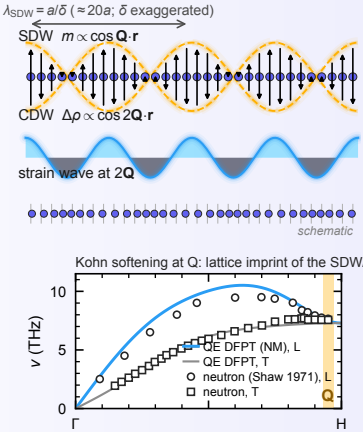
TOPOLOGY

(c) Fe-rich: ETT



MODULATE

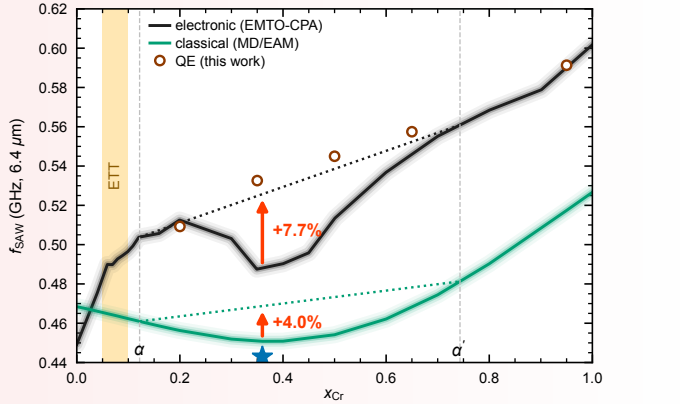
(d) SDW/CDW soften the lattice



STIFFEN

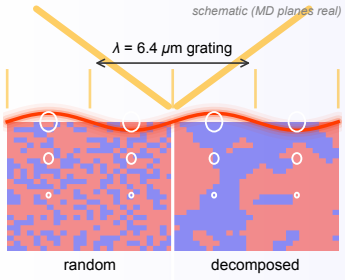
(e) electron → lattice: decomposition

$$C = C_{\text{ion}} + C_{\text{band}},\quad C_{\text{band}} \propto -\sum_k (\partial\varepsilon_k/\partial\eta)^2\,\delta(\varepsilon_k - E_F)$$



PROPAGATE

(f) Rayleigh wave in TGS



$C_{ij}\uparrow \Rightarrow G\uparrow \Rightarrow v_R = \eta(\nu)\sqrt{G/\rho}\uparrow \Rightarrow f = v_R/\lambda\uparrow$

Fe−36Cr, MD: 451 → 474 MHz (+5.1%)

electronic lever rule: +7.7%

DETECT

(g) (P)SAW vs pristine CASS

